

CS566 Deep Learning Term Project

Attention-Augmented Scale Hyperprior for Learned Image Compression

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Spring 2026

Main takeaway: the SE blocks were easy to integrate and lightweight, but they did not produce a meaningful Kodak rate-distortion gain.

Problem & Motivation

Problem

Images need to be compressed to a low bitrate *without* visibly damaging reconstruction quality.

Why Learned Compression

Neural codecs can learn the encoder, latent prior, and decoder jointly for rate-distortion optimization.

Project Objective

Keep the Ballé scale hyperprior fixed and test whether lightweight SE attention improves compression.

Fair Experimental Question

Compare a fine-tuned vanilla baseline (A_{ft}) against two SE variants: encoder-only (B) and encoder+decoder (C), using **true coded bitrate** on Kodak.

Dataset & Experimental Setup

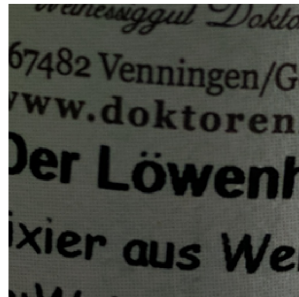
Final Setup

- ▶ CLIC train/val: 1,633 / 102
- ▶ Kodak test: 24 images
- ▶ 256×256 crops; 4 λ values

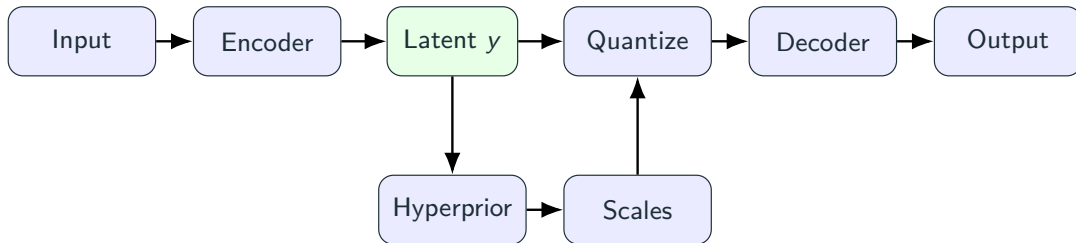
Example CLIC image



256x256 training patch



Method Overview

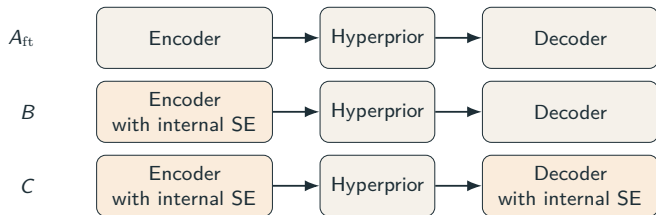


What Changes and What Stays Fixed

Changed: SE blocks are added in the transform network (encoder in B , encoder+decoder in C).

Fixed: the hyperprior, scale prediction, quantization, and entropy model stay the same.

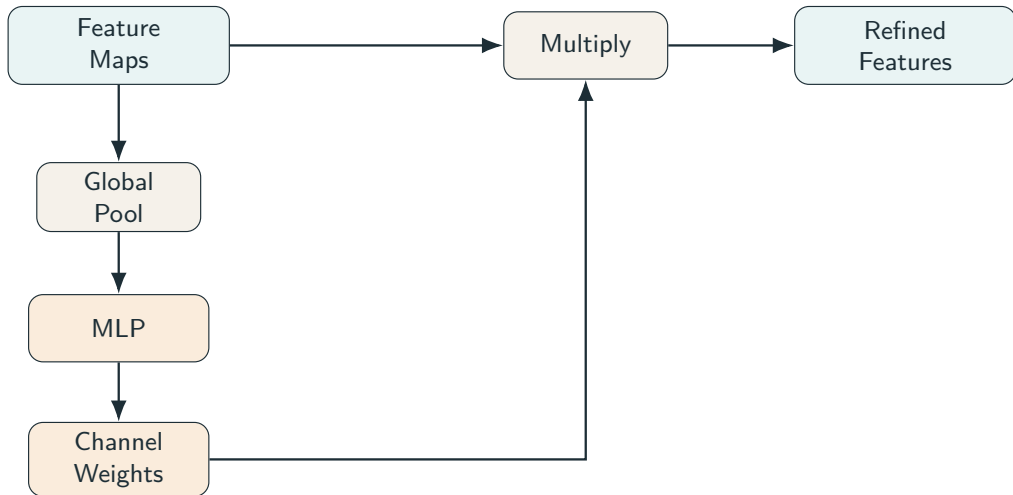
Model Architecture



Implemented Details

- ▶ B : SE after encoder conv 2
- ▶ C : + decoder SE after deconv 2
- ▶ Pretrained transfer;
+2.2k/ + 4.4k params

SE Block: General Idea



Training Objective

Rate-Distortion Loss

$$\mathcal{L} = \lambda D(x, \hat{x}) + R(\hat{y}) + R(\hat{z})$$

Training Setup

4 operating points

Adam: 10^{-4} backbone, 10^{-3} SE

3 freeze modes for SE adapter test

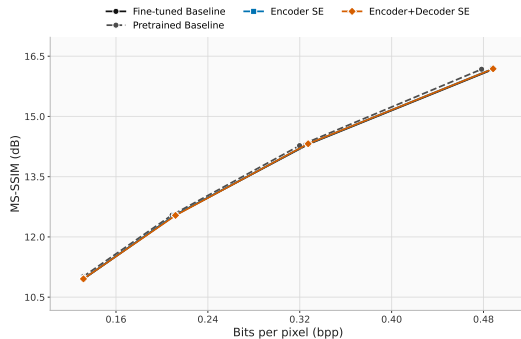
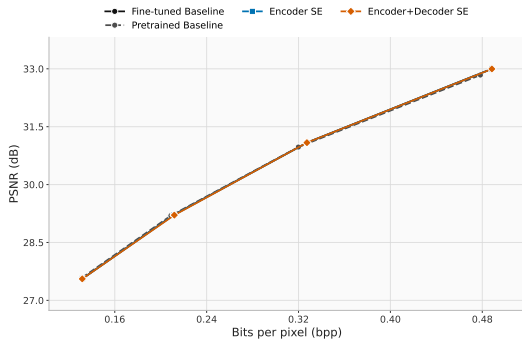
Evaluation

PSNR

true coded bitrate

measured with codec calls

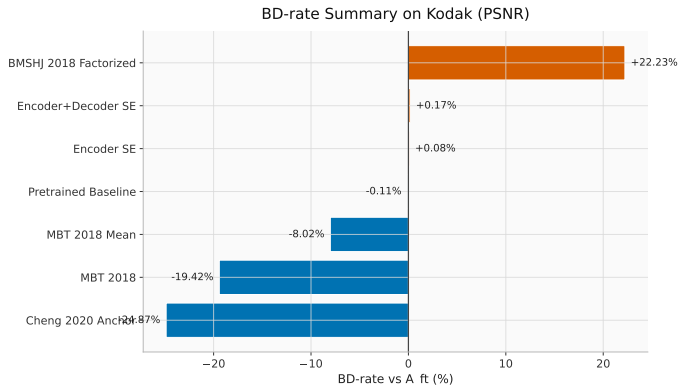
Results: RD Curves



Takeaway

project curves nearly overlap

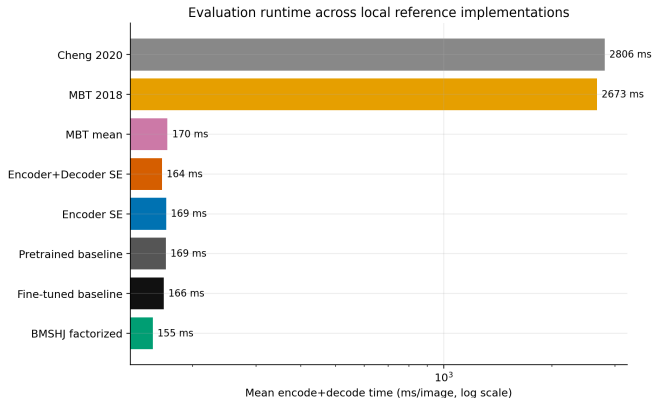
Results: BD-Rate Summary



Takeaway

- ▶ *B*: +0.08%
- ▶ *C*: +0.17%
- ▶ Stronger priors: -8% to -25%

Runtime Evaluation



Main Variant Runtime

Variant	Params (M)	Fwd (ms)	Codec (ms)
A_{ft}	5.1	11.4	166
B	5.1	11.6	169
C	5.1	11.7	164

Interpretation

- ▶ SE overhead: +0.043% (B) and +0.086% (C).
- ▶ Project variants stay within 164 to 169 ms/image.
- ▶ MBT 2018 / Cheng 2020 are $16.1\times$ / $16.9\times$ slower.

Image Comparison

Input



Baseline FT



Encoder SE



Encoder+Decoder SE



no visible difference at quality level 3

Discussion

Why SE Did Not Help Much Here

- ▶ No visual difference: all models look identical at quality level 3.
- ▶ The global blanket: one channel weight scales every pixel in that channel.
- ▶ The background bottleneck: flat background regions get scaled up too, which can waste bits.
- ▶ Cheng 2020 uses a 3D attention map, so flat areas and detailed regions can be weighted differently.

Conclusion

1. Implemented

Two SE-augmented scale-hyperprior variants, pretrained weight transfer, fair baselines, ablations, and cleaned evaluation.

2. Main Result

On Kodak, the SE variants do **not** show a meaningful rate-distortion gain over the fine-tuned vanilla baseline.

3. Takeaway

Lightweight transform-side attention is easy to add, but stronger priors and context models are the more promising direction.

If I Continued This Project

Run multiple seeds, and test richer attention placement or stronger entropy models.